The Fractional Riccati–Müntz–Padé Theorem

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May 18, 2026

Abstract

The classical theory of Padé approximation achieves its deepest results when the function being approximated is meromorphic outside a compact set of minimal logarithmic capacity—Stahl's compact. The present paper extends this circle of ideas to solutions of fractional Riccati equations. Specifically, for the equation

$$D^{\mu} y = P(u) + Q(u)\, y + R(u)\, y^2 \tag{1}$$

with polynomial coefficients $P, Q, R \in \mathbb{C}[u]$, order $\mu \in (0, 1]$, and homogeneous Riemann–Liouville initial condition, the unique solution $y(t, u)$ equals the limit, in logarithmic capacity, of its near-diagonal Müntz–Padé approximants of type $[M-1/M]$ in the variable $z = t^{\mu}$.

The argument turns on three interlocking observations. First, the solution admits an expansion $y = \sum_k a(k, u)\, t^{(k+1)\mu}$ whose coefficients obey an explicit Müntz–Tau recurrence driven by the Fox–Wright weights $\psi_k = \Gamma(k\mu+1)/\Gamma(k\mu+\mu+1)$. Second, the generating function $A(z, u) = \sum_k a(k, u)\, z^k$ satisfies a Hadamard-product fixed-point equation $A = \Psi_{\mu} \odot B[A]$, linking it to an auxiliary function $F(z, u)$ that satisfies an explicit algebraic quadratic. Third, a Mellin–Barnes integral representation, valid under a Gevrey-1 bound established here from an explicit Cauchy-integral interpolation, transfers meromorphic continuation from the pole structure of Ψ_{μ} to A, and a self-consistent fixed-point argument places K_A inside K_F.

The Padé denominators are identified as the orthogonal polynomials of the moment functional $\mathcal{L}[x^k] = a(k, u)$, computed by the Chebyshev–Wheeler algorithm, and convergence follows from a direct proof of Stahl's theorem for the generic two-valued algebraic class containing F, given in the appendix.

1 Setup and Notation

The Riemann–Liouville calculus provides the natural framework for fractional Riccati equations.

Definition 1 (Riemann–Liouville operators)**.** *For $\mu \in (0, 1]$ and f locally integrable on $(0, \infty)$,*

$$I^{\mu} f(t) := \frac{1}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} f(s)\, ds, \tag{2}$$

$$D^{\mu} f(t) := \frac{d}{dt}\big(I^{1-\mu} f\big)(t). \tag{3}$$

Lemma 1 (Power rule)**.** *For $s > -1$,*

$$I^{\mu} t^{s} = \frac{\Gamma(s+1)}{\Gamma(s+\mu+1)}\, t^{s+\mu}. \tag{4}$$

Proof. The substitution $\sigma = t\tau$ in the defining integral (2) reduces the integral to the beta function

$$B(\mu, s+1) = \frac{\Gamma(\mu)\Gamma(s+1)}{\Gamma(s+\mu+1)}, \tag{5}$$

which gives (4). □

Definition 2 (Fractional Riccati equation)**.**

$$D^\mu y(t,u) = P(u) + Q(u)\,y(t,u) + R(u)\,y(t,u)^2, \qquad I^{1-\mu}y(0,u) = 0. \tag{6}$$

Lemma 2 (Volterra form)**.** *Equation* (6) *is equivalent to*

$$y = I^\mu\big[P + Q\,y + R\,y^2\big]. \tag{7}$$

Proof. Application of I^μ to (6), the composition formula [1, Theorem 2.4], and the homogeneous initial condition $I^{1-\mu}y(0,u) = 0$ together kill the boundary term and yield (7). □

2 The Fox–Wright Weights

Definition 3 (Fox–Wright weights and kernel)**.**

$$\psi_k := \frac{\Gamma(k\mu+1)}{\Gamma(k\mu+\mu+1)}, \quad k \geq 0, \tag{8}$$

$$\Psi_\mu(z) := \sum_{k=0}^{\infty} \psi_k\, z^k. \tag{9}$$

Lemma 3 (Eigenvalue action)**.** *For* $k \geq 0$*,*

$$I^\mu t^{(k+1)\mu} = \psi_{k+1}\, t^{(k+2)\mu}. \tag{10}$$

Proof. Application of Lemma 1, equation (4), with $s = (k+1)\mu$ gives (10). □

Lemma 4 (Radius of convergence)**.** Ψ_μ *has radius of convergence* 1*.*

Proof. By Stirling,

$$\psi_k = \frac{\Gamma(k\mu+1)}{\Gamma(k\mu+\mu+1)} \sim (k\mu)^{-\mu} \qquad (k\to\infty). \tag{11}$$

Hence

$$\frac{1}{k}\log|\psi_k| = -\frac{\mu}{k}\,\log(k\mu) + o\big(\tfrac{1}{k}\big) \;\longrightarrow\; 0, \tag{12}$$

so $\limsup_{k\to\infty} |\psi_k|^{1/k} = 1$, and the radius of convergence of Ψ_μ is 1. □

3 The Müntz–Tau Series

Theorem 1 (Müntz–Tau recurrence)**.** *The unique solution of* (7) *admits the expansion*

$$y(t,u) = \sum_{k\geq 0} a(k,u)\, t^{(k+1)\mu} \tag{13}$$

on $[0, T_0]$ *for some* $T_0 > 0$*, with*

$$a(0,u) = P(u)\,\psi_0, \tag{14}$$

$$a(k,u) = \psi_k \left(Q(u)\,a(k-1,u) + R(u) \sum_{j+\ell=k-1} a(j,u)\,a(\ell,u) \right), \quad k \geq 1. \tag{15}$$

Proof. The map

$$\mathcal{T}[y] := I^\mu(P + Qy + Ry^2) \tag{16}$$

is a contraction on the ball $\{y \in C[0, T_0] : \|y\|_\infty \le 2|P(u)|\, T_0^\mu/\Gamma(\mu+1)\}$ whenever

$$T_0^\mu \;<\; \frac{\Gamma(\mu+1)}{|Q(u)| + 2|R(u)|\,\|y\|_\infty}, \tag{17}$$

making the local existence radius explicit and u-dependent. Substitution of the Müntz ansatz (13) and application of Lemma 3 match coefficients of $t^{(k+1)\mu}$: $k = 0$ gives (14); $k \ge 1$ gives (15). $\square$

4 The Algebraic Quadratic via Generating Functions

Conventions. Coefficient extraction

$$[z^k]b := b_k, \tag{18}$$

Cauchy product

$$[z^k](F \cdot G) = \sum_j f_j g_{k-j}, \tag{19}$$

Hadamard product

$$[z^k](F \odot G) = f_k g_k. \tag{20}$$

Definition 4 (Auxiliary generating function)**.** *Set $f(0, u) := P\psi_0$ and, for $k \ge 1$,*

$$f(k, u) := Q\, f(k-1, u) + R \sum_{j+\ell=k-1} f(j, u) f(\ell, u) + P\psi_k. \tag{21}$$

Define

$$F(z, u) := \sum_{k\ge 0} f(k, u)\, z^k. \tag{22}$$

Remark 1. *In (15) the bracket is premultiplied by ψ_k; in (21) the term $P\psi_k$ is a separate source. Hence $F \ne A$ and neither $\Psi_\mu \cdot A$ nor $\Psi_\mu \odot A$ equals F. The bridge is the Hadamard fixed point of Section 7. Comparison of first-order coefficients gives*

$$a(1, u) = \psi_1\big(QP\psi_0 + R(P\psi_0)^2\big), \tag{23}$$

$$f(1, u) = QP\psi_0 + R(P\psi_0)^2 + P\psi_1, \tag{24}$$

which differ as elements of $\mathbb{C}[u]$ for generic P, Q, R, μ.

Theorem 2 (Algebraic quadratic for F)**.** *F satisfies*

$$zR\, F^2 - (1 - zQ)\, F + P\, \Psi_\mu = 0, \tag{25}$$

with the closed form

$$F(z, u) = \frac{(1 - zQ) - \sqrt{(1 - zQ)^2 - 4zPR\, \Psi_\mu(z)}}{2zR}. \tag{26}$$

Proof. Coefficient-by-coefficient,

$$G := zRF^2 - (1 - zQ)F + P\Psi_\mu \tag{27}$$

satisfies $[z^0]G = 0$ and $[z^k]G = 0$ for $k \ge 1$ by (21). Expansion of the discriminant,

$$\sqrt{\Delta} = 1 - z(Q + 2PR\psi_0) + O(z^2), \tag{28}$$

shows the minus branch of (26) has constant term $P\psi_0 = f(0, u)$, the unique formal power-series root since

$$\partial_F\big[zRF^2 - (1 - zQ)F + P\Psi_\mu\big]\big|_{z=0} = -1. \tag{29}$$

$\square$

5 The Chebyshev–Wheeler Algorithm

5.1 Points of growth and positive-definiteness

Definition 5. *For λ non-decreasing right-continuous, x_0 is a point of growth if $\lambda(x_0+\varepsilon)-\lambda(x_0-\varepsilon) > 0$ for all $\varepsilon > 0$. Set*

$$G(\lambda) := \operatorname{supp}(d\lambda), \qquad N_\lambda := |G(\lambda)|. \tag{30}$$

Lemma 5. *The form*

$$\langle f, g\rangle_\lambda := \int f g \, d\lambda \tag{31}$$

is positive-definite on $\mathbb{R}[x]$ iff $N_\lambda = \infty$.

Proof. ($\Leftarrow$) A degree-d polynomial vanishes at $\leq d$ points, so f^2 is positive on some point of growth. ($\Rightarrow$) If $G(\lambda) = \{x_1, \ldots, x_M\}$, then

$$f(x) := \prod_{j=1}^{M} (x - x_j) \tag{32}$$

vanishes on $G(\lambda)$, giving $\int f^2 \, d\lambda = 0$. □

5.2 Wheeler's algorithm

Auxiliary array. Initial data and recurrence:

$$\sigma(-1, \ell) = 0, \tag{33}$$

$$\sigma(0, \ell) = \mu(\ell), \tag{34}$$

$$\sigma(k, \ell) = \sigma(k-1, \ell+1) - \alpha(k-1)\sigma(k-1, \ell) - \beta(k-1)\sigma(k-2, \ell). \tag{35}$$

Coefficients.

$$\alpha(0) = \mu(1)/\mu(0), \qquad \beta(0) = \mu(0), \tag{36}$$

$$\alpha(k) = \frac{\sigma(k, k+1)}{\sigma(k, k)} - \frac{\sigma(k-1, k)}{\sigma(k-1, k-1)}, \tag{37}$$

$$\beta(k) = \frac{\sigma(k, k)}{\sigma(k-1, k-1)}. \tag{38}$$

Orthogonal polynomials.

$$\pi_{-1} = 0, \qquad \pi_0 = 1, \tag{39}$$

$$\pi_{k+1} = (x - \alpha(k))\, \pi_k - \beta(k)\, \pi_{k-1}. \tag{40}$$

Theorem 3 (Wheeler)**.** *$\sigma(k, k) > 0$ and*

$$\int \pi_j \pi_k \, d\lambda = \delta_{jk} \prod_{i=0}^{k} \beta(i). \tag{41}$$

Proof. Set

$$\tilde{\sigma}(k, \ell) := \int \pi_k \, x^\ell \, d\lambda. \tag{42}$$

This matches σ in initial data and recurrence, so they coincide. Induction shows $\pi_k \perp \{1, \ldots, x^{k-1}\}$, giving $\sigma(k, \ell) = 0$ for $\ell < k$ and $\sigma(k, k) = \langle \pi_k, \pi_k\rangle_\lambda > 0$ by Lemma 5. The norm formula follows from

$$\langle \pi_k, \pi_k\rangle = \beta(k)\, \langle \pi_{k-1}, \pi_{k-1}\rangle. \tag{43}$$

□

5.3 Symbolic transfer

Definition 6. $\{a(k,u)\}$ *is* quasi-definite *if the Hankel determinants*

$$H_n(u) := \det\bigl(a(i+j,u)\bigr)_{0\le i,j<n} \tag{44}$$

satisfy $H_n(u) \not\equiv 0$ *for all* n.

Theorem 4 (Symbolic Wheeler)**.** *Under quasi-definiteness,* $\sigma(k,k,u) \not\equiv 0$*, and the orthogonal polynomials* P_k *of the functional* $\mathcal{L}[x^\ell] = a(\ell,u)$ *satisfy*

$$\mathcal{L}[P_j P_k] = \delta_{jk} \prod_{i=0}^{k} \beta_i(u) \quad \text{in } \mathbb{C}(u). \tag{45}$$

Proof. The Hankel ratio gives

$$\sigma(k,k,u) = \frac{H_{k+1}(u)}{H_k(u)} \not\equiv 0. \tag{46}$$

The identity

$$\sigma(k,\ell,u) = \mathcal{L}[x^\ell P_k] \tag{47}$$

holds by induction, and $\mathcal{L}[P_k^2] = \sigma(k,k,u)$ telescopes to (45). □

6 Padé Denominators via Gragg's Theorem

Padé convention. The M-th approximant has $\deg N_M \le M-1$, $\deg D_M \le M$, and is therefore of type $[M-1/M]$. Its denominator sequence $\{D_M\}$ is identical to that of the diagonal $[M/M]$ table; these approximants are *diagonal*, following Stahl.

Theorem 5 (Gragg, symbolic form)**.** *There is a unique pair* (N_M, D_M) *with* $\deg N_M \le M-1$, $\deg D_M \le M$*,* $D_M(0,u) = 1$*, and*

$$D_M\, A - N_M = O(z^{2M}), \tag{48}$$

satisfying

$$D_M(z,u) = z^M\, P_M(z^{-1},u). \tag{49}$$

Proof. The M matching conditions

$$[z^j](D_M A) = 0, \qquad j = M,\dots,2M-1, \tag{50}$$

become, with $Q_M(x,u) := x^M D_M(x^{-1},u)$, the orthogonality

$$\mathcal{L}[x^i Q_M] = 0, \qquad i = 0,\dots,M-1. \tag{51}$$

The monic Q_M of degree $\le M$ matching these conditions is unique by Theorem 4: $Q_M = P_M$. Since P_M is monic, $D_M(0,u) = 1$. □

The truncated Jacobi matrix.

$$J_M(u) := \begin{pmatrix} \alpha(0,u) & 1 & & \\ \beta(1,u) & \alpha(1,u) & 1 & \\ & \ddots & \ddots & \ddots \\ & & \beta(M-1,u) & \alpha(M-1,u) \end{pmatrix}. \tag{52}$$

On the orthonormal basis $\widehat{\pi}_k = \pi_k / \sqrt{\prod_{i=0}^{k} \beta(i,u)}$, the three-term recurrence (40) gives the defect equation

$$J_M(u)\, v(\lambda) = \lambda\, v(\lambda) - \pi_M(\lambda,u)\, e_M, \qquad v(\lambda) = \bigl(\widehat{\pi}_0(\lambda),\dots,\widehat{\pi}_{M-1}(\lambda)\bigr)^\top. \tag{53}$$

Theorem 6 (Spectral identification)**.** *Under quasi-definiteness:*

1. *The eigenvalues of $J_M(u)$ are exactly the zeros of $\pi_M(\cdot, u)$, hence the reciprocals of the zeros of $D_M(z,u) = z^M \pi_M(z^{-1}, u)$ (Theorem 5).*
2. *For each eigenvalue λ the right eigenvector is $v(\lambda) = (\widehat{\pi}_0(\lambda), \ldots, \widehat{\pi}_{M-1}(\lambda))^\top$, and the defect equation* (53) *holds.*
3. *As $M \to \infty$, the reciprocal empirical eigenvalue measure $\nu_M^{\mathrm{rec}}(u) := M^{-1} \sum_{\lambda \in \operatorname{spec} J_M(u)} \delta_{1/\lambda}$ converges to the pushforward $(1/z)_* \, \nu_{K_A(u)}$ of the equilibrium measure of the Stahl compact $K_A(u)$, weakly on compacta of $\widehat{\mathbb{C}} \setminus \{0\}$.*

Proof. (1) By the determinantal identity

$$\widehat{\pi}_k(\lambda, u) = \det\bigl(\lambda I_k - \widetilde{J}_k(u)\bigr) \Big/ \prod_{i=2}^{k} \sqrt{\beta(i,u)}, \tag{54}$$

with $\widetilde{J}_k(u)$ the symmetric tridiagonal form of $J_k(u)$, the zeros of $\pi_M(\cdot, u)$ coincide with $\operatorname{spec} J_M(u)$; the identification $D_M(z,u) = z^M \pi_M(z^{-1}, u)$ gives the reciprocal relationship. (2) Substituting $v(\lambda)$ into $J_M(u)$ at rows $k < M-1$ yields $(J_M(u) v(\lambda))_k = \lambda \, \widehat{\pi}_k(\lambda)$ by (40); at row $M-1$ the absent $\pi_M(\lambda)$ entry contributes $-\pi_M(\lambda, u) \, e_M$, which is (53). (3) The zeros of D_M are reciprocals of $\operatorname{spec} J_M(u)$; Theorem 11 and (121) give weak-$*$ convergence of the zero-counting measure of D_M to $\nu_{K_A(u)}$, and the pushforward under the homeomorphism $z \mapsto 1/z$ of $\widehat{\mathbb{C}} \setminus \{0\}$ yields the stated reciprocal limit on compacta avoiding the origin. □

Remark 2 (Spectral measure, constant-coefficient case)**.** *When $P(u), Q(u), R(u) \in \mathbb{C}$ are constants, A is meromorphic on $\mathbb{C}$ with simple poles $\mathcal{P}_A$, and the spectral measure $\mu_A := \sum_{z^* \in \mathcal{P}_A} \operatorname{Res}(A, z^*) \, \delta_{z^*}$ has moments $\mathcal{L}[x^k] = a(k,u)$. In this setting $J_M(u)$ is the truncated Jacobi operator of μ_A (Stone's theorem) and N_M / D_M is the rank-M Stieltjes transform of μ_A.*

7 Meromorphic Continuation of A

This section establishes that A is meromorphic on $\widetilde{X} \setminus K_A(u)$, the universal cover of $\mathbb{C} \setminus \{0\}$ minus a compact $K_A(u) \subseteq K_F(u)$ of minimal logarithmic capacity.

7.1 Hadamard fixed-point equation

Lemma 6. *With*

$$B[A] := P + zQA + zRA^2, \tag{55}$$

the identity

$$A = \Psi_\mu \odot B[A] \tag{56}$$

holds.

Proof. $[z^0]B[A] = P$ gives $\psi_0 P = a(0)$. For $k \geq 1$,

$$[z^k]B[A] = Q \, a(k-1) + R \sum_{j+\ell=k-1} a(j) \, a(\ell), \tag{57}$$

and multiplication of (57) by ψ_k recovers $a(k)$ by (15). □

7.2 Mellin transform of Ψ_μ

Lemma 7 (Mellin transform)**.** *For $0 < \Re s < 1$,*

$$\int_0^\infty \Psi_\mu(-x)\, x^{s-1}\, dx = \Gamma(s)\Gamma(1-s)\frac{\Gamma(1-s\mu)}{\Gamma(\mu+1-s\mu)} =: M_{\Psi_\mu}(s). \tag{58}$$

M_{Ψ_μ} extends meromorphically to $\mathbb{C}$ with poles at $\mathbb{Z}_{\leq 0}$, $\mathbb{Z}_{\geq 1}$, and $(1+m)/\mu$ for $m \geq 0$. A pole at $(1+m)/\mu$ is cancelled by a denominator pole at $1+(1+n)/\mu$ iff $m-n=\mu$; for $\mu \in (0,1]$ this requires $\mu = 1$. For $\mu \in (0,1)$ the lattice $\{(1+m)/\mu\}$ survives uncancelled. At $\mu = 1$ the collapse is explicit:

$$M_{\Psi_1}(s) = \Gamma(s)\,\Gamma(1-s)\,\frac{\Gamma(1-s)}{\Gamma(2-s)} = \frac{\Gamma(s)\,\Gamma(1-s)}{1-s}, \tag{59}$$

whose order-two pole at $s = 1$ is precisely what produces the logarithmic kernel $\Psi_1(z) = -\log(1-z)/z$: a simple Mellin pole would give a power law, the double pole gives the log*.*

Proof. Write

$$\Psi_\mu(-x) = \sum_{k\geq 0} \frac{(-x)^k\,\varphi(k)}{k!}, \qquad \varphi(k) := k!\,\psi_k. \tag{60}$$

The interpolation

$$\varphi(s) = \frac{\Gamma(s+1)\,\Gamma(s\mu+1)}{\Gamma(s\mu+\mu+1)} \tag{61}$$

is entire. Stirling on a vertical line $\Re s = \sigma$ gives

$$|\varphi(\sigma+i\tau)| \sim C(\sigma,\mu)\,|\tau|^{\,\sigma+\mu\sigma-\mu+1/2}\,e^{-(\pi/2)|\tau|} \qquad (|\tau| \to \infty), \tag{62}$$

where $C(\sigma,\mu) > 0$ depends only on σ and $\mu \in (0,1]$. The exponential rate $\pi/2 < \pi$ satisfies Hardy's growth condition [8, Lecture XI, §§11.2, 11.10–11.13], and Ramanujan's master theorem gives

$$\int_0^\infty \Psi_\mu(-x)\,x^{s-1}\,dx = \Gamma(s)\,\varphi(-s). \tag{63}$$

The cancellation analysis is recorded in the statement. □

7.3 Mellin–Barnes realisation

Lemma 8. *Let*

$$G(z,u) = \sum_{k\geq 0} g_k(u)\,z^k \tag{64}$$

be holomorphic in $|z| < \rho$, with Ramanujan-normalised entire interpolation

$$\widehat{G}_R(s,u) := \frac{\widehat{G}(s,u)}{\Gamma(s+1)}, \tag{65}$$

where $\widehat{G}$ is the Cauchy interpolation of (85) below, satisfying $\widehat{G}_R(k,u) = g_k(u)$ at integers and the Gevrey-1 bound

$$|\widehat{G}(\sigma+i\tau,u)| \leq K'_G\,C_G^{|\sigma|+1}\,|\Gamma(\sigma+i\tau+1)|. \tag{66}$$

Fix $c \in (0,1)$ avoiding poles of M_{Ψ_μ}. For $|\arg(-z)| < \pi/2$ and $|z| < \min(1,\rho)$,

$$(\Psi_\mu \odot G)(z,u) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} M_{\Psi_\mu}(s)\,\widehat{G}_R(-s,u)\,(-z)^{-s}\,ds. \tag{67}$$

The integrand decays as

$$e^{-(\pi/2)|\tau|}\,|\tau|^{C_1+C_2}, \tag{68}$$

so the integral converges absolutely on the sector and continues meromorphically to $\widetilde{X}$ minus the projected pole set of $M_{\Psi_\mu}\widehat{G}_R$.

Proof. Closing the contour to the left, the simple pole of $\Gamma(s)$ at $s = -k$ has residue $(-1)^k/k!$. Split

$$M_{\Psi_\mu}(s) = \Gamma(s) \cdot H(s), \qquad H(s) := \frac{\Gamma(1-s)\,\Gamma(1-s\mu)}{\Gamma(\mu+1-s\mu)}, \tag{69}$$

so that $\Gamma(s)$ is the only factor producing a pole at $s = -k$. At $s = -k$,

$$H(-k) = \Gamma(1+k)\,\frac{\Gamma(1+k\mu)}{\Gamma(\mu+1+k\mu)} = k!\,\psi_k, \tag{70}$$

$$\widehat{G}_R(-s,u)\big|_{s=-k} = \widehat{G}_R(k,u) = g_k(u), \tag{71}$$

$$(-z)^{-s}\big|_{s=-k} = (-1)^k z^k. \tag{72}$$

Therefore

$$\operatorname*{Res}_{s=-k}\Big[M_{\Psi_\mu}(s)\,\widehat{G}_R(-s,u)\,(-z)^{-s}\Big] = \frac{(-1)^k}{k!}\cdot k!\,\psi_k \cdot g_k(u)\cdot(-1)^k z^k = \psi_k\, g_k(u)\, z^k, \tag{73}$$

using $(-1)^k(-1)^k = 1$. Summation over $k \ge 0$ recovers

$$\sum_{k\ge 0} \psi_k\, g_k\, z^k = \Psi_\mu \odot G. \tag{74}$$

Continuation to $\widetilde{X}$ is by the identity principle along chains of overlapping disks. □

7.4 Stahl compact for F and for A

Lemma 9. *There is a unique compact $K_F(u) \supseteq K_\Psi$ of minimal capacity outside which $\sqrt{\Delta_F(z,u)}$, hence F, is single-valued meromorphic on $\widetilde{X}$. Here*

$$\Delta_F(z,u) = (1 - zQ(u))^2 - 4zP(u)R(u)\,\Psi_\mu(z), \tag{75}$$

and K_Ψ is the projected pole set of M_{Ψ_μ}.

Proof. By Lemma 7, $\Delta_F(z,u)$ is meromorphic on $\widetilde{X} \setminus K_\Psi$. Theorem 10 of the appendix produces the unique minimal-capacity compact $K_F(u) \supseteq K_\Psi$ outside which $\sqrt{\Delta_F(z,u)}$ is single-valued. □

Theorem 7 (Meromorphic continuation of A)**.** *Under quasi-definiteness, A is meromorphic on $\widetilde{X} \setminus K_A(u)$ with $K_A(u) \subseteq K_F(u)$.*

Proof. **Step 1: Gevrey-1 bound on $\{a(k,u)\}$ and $\{g_k(u)\}$.** Fix u in a compact subset of parameter space. Since ψ_k is decreasing in k for fixed $\mu \in (0,1]$, the supremum is attained at $k = 0$:

$$C_0 := \sup_{k\ge 0} |\psi_k| = \psi_0 = \frac{1}{\Gamma(\mu+1)} < \infty \quad \text{(by Lemma 4).} \tag{76}$$

Set

$$M_P := |P(u)|, \qquad M_Q := |Q(u)|, \qquad M_R := |R(u)|, \tag{77}$$

and define the (local) Gevrey constants

$$K_G := C_0\,\max(M_P,\,1), \qquad C_G := C_0\,(M_Q + 2K_G\,M_R) + 1. \tag{78}$$

The bound to be established is

$$|a(k,u)| \le K_G\,C_G^k\,k! \qquad \text{for all } k \ge 0. \tag{79}$$

The base case $k = 0$ uses $|a(0,u)| = |P\psi_0| \le C_0 M_P \le K_G \le K_G\, C_G^0\, 0!$. Inductively, assuming (79) through $k-1$,

$$|a(k,u)| \le C_0\big(M_Q\, |a(k-1)| + M_R \sum_{j+\ell=k-1} |a(j)|\, |a(\ell)|\big). \tag{80}$$

The convolution sum is bounded using

$$\sum_{j+\ell=n} j!\, \ell! \le (n+1)!, \tag{81}$$

giving $\sum_{j+\ell=k-1} K_G^2\, C_G^{k-1}\, j!\, \ell! \le K_G^2\, C_G^{k-1}\, k!$. Hence

$$|a(k)| \le C_0 M_Q\, K_G\, C_G^{k-1}(k-1)! + C_0 M_R\, K_G^2\, C_G^{k-1} k! \le K_G\, C_G^k\, k! \cdot \frac{C_0(M_Q + K_G\, M_R)}{C_G} \le K_G\, C_G^k\, k!, \tag{82}$$

since $C_0(M_Q + K_G\, M_R)/C_G < 1$ by (78). The bound for $\{g_k(u)\}$ follows from

$$g_k = Q\, a(k-1) + R \sum_{j+\ell=k-1} a(j)\, a(\ell) + P\, [k=0], \tag{83}$$

yielding

$$|g_k(u)| \le K_G''\, C_G^{k+1}\, k!, \qquad K_G'' := K_G(M_Q + K_G\, M_R)/C_G^2. \tag{84}$$

Step 2: Cauchy-integral interpolation. Fix $0 < r < \rho_0$ where G is holomorphic. Define

$$\widehat{G}(s,u) := \Gamma(s+1) \cdot \frac{1}{2\pi i} \oint_{|z|=r} \frac{G(z,u)}{z^{s+1}}\, dz, \qquad \widehat{G}_R(s,u) := \frac{\widehat{G}(s,u)}{\Gamma(s+1)}. \tag{85}$$

At integer points,

$$\widehat{G}(k,u) = k!\, g_k(u), \qquad \widehat{G}_R(k,u) = g_k(u), \tag{86}$$

the latter being the Ramanujan normalisation used in Lemma 8. The Cauchy-kernel factor

$$\kappa(s,u) := \frac{1}{2\pi i} \oint_{|z|=r} \frac{G(z,u)}{z^{s+1}}\, dz \tag{87}$$

is entire in s of exponential type at most $\log(1/r)$ along the imaginary axis. Multiplication by $\Gamma(s+1)$ does not alter this type. Carlson's theorem requires

$$\log(1/r) < \pi \quad \Longleftrightarrow \quad r > e^{-\pi}. \tag{88}$$

Since G is holomorphic on $|z| < \rho_0$, $\kappa(s,u)$ is independent of r in $(0, \rho_0)$, so r may be enlarged to any value in $(e^{-\pi}, \rho_0)$, which is nonempty provided $\rho_0 > e^{-\pi}$. If $\rho_0 \le e^{-\pi}$, fix any $r_0 \in (0, \rho_0)$ and replace z by λz for a scalar λ with

$$\lambda r_0 \in (e^{-\pi},\, 1); \tag{89}$$

this rescaling reduces to the previous case and is undone after the Mellin–Barnes representation by the inverse scaling. Carlson's theorem then gives uniqueness of $\widehat{G}$ as the entire interpolation of $\{k!\, g_k(u)\}_{k\ge 0}$ of exponential type $< \pi$. The Cauchy integral satisfies

$$|\kappa(s,u)| \le \frac{M(r,u)}{r^{\Re s}}, \qquad M(r,u) := \sup_{|z|=r} |G(z,u)|, \tag{90}$$

which gives

$$|\widehat{G}(\sigma + i\tau, u)| \le \frac{M(r,u)\, |\Gamma(\sigma + i\tau + 1)|}{r^\sigma}. \tag{91}$$

Combination of (91) with the integer-point Gevrey bound (84) and the Phragmén–Lindelöf principle in each vertical strip (applied to $\widehat{G}(s,u)/[K'_G C_G^{|\sigma|+1}\Gamma(s+1)]$, which is bounded on the imaginary axis and on $\Re s = N$ for each $N \in \mathbb{Z}_{\geq 0}$ by the Gevrey bound and Stirling, and of finite order in the strip) yields (66) with

$$K'_G := M(r,u)/r, \tag{92}$$

and C_G as in (78) (after enlarging C_G to absorb $1/r$).

Step 3: Mellin–Barnes representation. By Lemma 6,

$$A = \Psi_\mu \odot G, \qquad G = B[A]. \tag{93}$$

Lemma 8 applies and yields, for $|\arg(-z)| < \pi/2$ and $|z| < \min(1, \rho_0)$, the integral representation (67), holomorphic on the open sector. By the identity principle, A extends meromorphically to

$$\widetilde{X} \setminus (K_\Psi \cup S_G), \tag{94}$$

where S_G is the singularity locus of $\widehat{G}_R$.

Step 4 (Branch locus). The formal series $A(z,u)$ is unique by Theorem 1 and satisfies $A(z,u) = \Psi_\mu(z) \odot B[A](z,u)$ by Lemma 6. For any compact $K \supseteq K_\Psi$ on which $A(\cdot,u)$ admits meromorphic continuation to $\widetilde{X} \setminus K$, Lemma 8 produces a meromorphic continuation of $\Psi_\mu \odot B[A]$ to $\widetilde{X} \setminus K$, and the two meromorphic functions agree on the disk of convergence, hence on $\widetilde{X} \setminus K$ by the identity principle. Lemma 13 shows the branch locus of $A(\cdot,u)$ lies in the branch locus of $F(\cdot,u)$ together with K_Ψ, so $A(\cdot,u)$ is meromorphic on $\widetilde{X} \setminus K_F(u)$. Existence of the minimal compact $K_A(u) \subseteq K_F(u)$ follows from Theorem 10. □

8 The Fractional Riccati–Müntz–Padé Theorem

Theorem 8 (Fractional Riccati–Müntz–Padé theorem)**.** *Under quasi-definiteness, the Müntz–Padé approximant*

$$y_M(t,u) := t^\mu \frac{N_M(t^\mu,u)}{D_M(t^\mu,u)} \tag{95}$$

satisfies

$$y(t,u) = \lim_{M\to\infty} y_M(t,u) \tag{96}$$

in logarithmic capacity on compact subsets of the t-universal-cover preimage of $\widetilde{X} \setminus K_A(u)$.

Proof. By Theorem 1,

$$y(t,u) = t^\mu A(t^\mu,u); \tag{97}$$

N_M/D_M is the $[M-1/M]$ Padé approximant of A by Theorem 5.

Equivalence of denominator sequences. D_M is determined by (51). The $[M/M]$ approximant adds one matching condition that fixes the leading coefficient of N_M, not the denominator. Hence $\{D_M\}_{M\geq 1}$ is identical for both types and Stahl's theorem applies to either. By quasi-definiteness the Padé table of $A(\cdot,u)$ is normal: all Hankel determinants $H_n(u) \neq 0$, and the $[M-1/M]$ and $[M/M]$ entries share the denominator $\pi_M(\cdot,u)$ (the Gragg connection, Theorem 5).

Convergence in capacity. By Theorem 7, A is meromorphic on $\widetilde{X} \setminus K_A(u)$. Theorem 11 of the appendix gives

$$\frac{N_M}{D_M} \longrightarrow A \quad \text{in capacity on compacta of } \widetilde{X} \setminus K_A(u). \tag{98}$$

Transfer to the t-plane. Fix the principal-branch sector

$$\Sigma := \{t \in \widetilde{\mathbb{C}}^* : \arg t \in (-\pi, \pi]\}, \tag{99}$$

on which $z = t^\mu$ is a biholomorphism onto its image. The map $t \mapsto t^\mu$ is a C^1 diffeomorphism on $\Sigma \setminus \{0\}$ with bounded distortion on compacta, and the classical capacity-transformation identity (cf. Ransford [9, Thm. 5.2.5]) gives

$$\operatorname{cap}(\{t \in \Sigma : t^\mu \in K\}) = \operatorname{cap}(K)^{1/\mu} \tag{100}$$

for every compact K in the image. On the full universal cover of $\mathbb{C}\setminus\{0\}$, the preimage decomposes into countably many such sectorial sheets, and convergence in capacity on each sheet transfers from the z-plane to the t-plane sheetwise. The statement of the theorem refers to this sheetwise transfer. □

Lemma 10 (No real poles)**.** *In the constant-coefficient case $P(u), Q(u), R(u) \in \mathbb{C}$ with $R(u) \neq 0$, the generating function $A(z, u)$ has no poles on $(0, \infty)$.*

Proof. A pole $z^*(u)$ of $A(\cdot, u)$ is inherited from a discriminant zero of $F(\cdot, u)$, i.e. $(1 - z^* Q(u))^2 = 4z^* P(u) R(u)\, \Psi_\mu(z^*)$. Expanding F from (26) about z^* with $\Delta_F(z, u) = \Delta'_F(z^*, u)\,(z - z^*) + O((z - z^*)^2)$ gives

$$\operatorname{Res}\big(F(\cdot, u), z^*\big) = \frac{\sqrt{-\Delta'_F(z^*, u)}}{2\, z^* R(u)}; \tag{101}$$

transferring to A by Lemma 13 and substituting $\Delta'_F(z^*, u)$ from the defining discriminant relation reduces this to

$$\operatorname{Res}\big(A(\cdot, u), z^*\big) = -\frac{(z^*)^{(\mu-1)/\mu}}{R(u)\, \mu}. \tag{102}$$

For $z^* \in (0, \infty)$ this residue is nonzero; but the Volterra fixed point $y(t, u) = t^\mu A(t^\mu, u)$ is holomorphic on every compact real t-interval on which the Müntz series (13) converges (Theorem 1), a contradiction. Hence $A(\cdot, u)$ has no poles on $(0, \infty)$. □

Corollary 1 (Uniform convergence on $\mathbb{R}_+$)**.** *In the constant-coefficient case, for every compact $[0, T] \subset \mathbb{R}_+$,*

$$\sup_{t \in [0,T]} \big|y(t) - y_M(t)\big| \longrightarrow 0 \qquad \text{as } M \to \infty. \tag{103}$$

Proof. By Lemma 10, $\delta := \operatorname{dist}\big([0, T], \mathcal{P}_A\big) > 0$, where $\mathcal{P}_A$ denotes the pole set of $A(\cdot, u)$. On $U_\delta := \{z \in \mathbb{C} : \operatorname{dist}(z, [0, T]) < \delta/2\}$, $A(\cdot, u)$ is holomorphic and bounded. Convergence in capacity on U_δ (Theorem 11) upgrades to uniform convergence on $[0, T]$: the capacity-zero exceptional sets cannot meet $[0, T]$ for all sufficiently large M, and a normal-families bound finishes the argument. The substitution $z = t^\mu$ transfers uniform convergence from $[0, T^\mu]$ to $[0, T]$. □

9 Exact Option Pricing from the Müntz–Padé Characteristic Function

Given the characteristic function in the form produced by the Müntz–Padé expansion, the European option price is computed *exactly*: by a finite sum of residues, with no quadrature, no contour truncation, no damping parameter, and no fast Fourier transform. We prove the pricing identity (Theorem 9), its error bound (Proposition 1), and its optimality (Remark 3).

Standing setup. Fix maturity T and rate $r = 0$; the general case differs only by the discount factor e^{-rT} and the forward shift $k \mapsto k - rT$. Let $X_T = \log(S_T/S_0)$ and write the order-M characteristic function as

$$\varphi_M(v) = \exp\big(\Phi_M(v)\big), \qquad \Phi_M(v) = \sum_{j=1}^{M} \frac{c_j}{v - \eta_j}, \tag{104}$$

the partial fraction of the rational cumulant exponent Φ_M in the Fourier variable v, with simple poles $\{\eta_j\}_{j=1}^{M}$ and residues c_j read off the orthogonal-polynomial recurrence (Theorem 5). The normalisations

$$\varphi_M(0) = 1, \qquad \varphi_M(-i) = \mathbb{E}[e^{X_T}] = 1 \tag{105}$$

hold *exactly for every* M, because the Riccati source $P(v) = \frac{1}{2}(-v^2 - iv)$ vanishes at $v = 0$ and $v = -i$, so every $a(k, 0) = a(k, -i) = 0$ and $\Phi_M(0) = \Phi_M(-i) = 0$. By Hermitian symmetry $\varphi_M(-\bar{v}) = \overline{\varphi_M(v)}$ and the no-real-poles property (Lemma 10, transported to the v-plane through the analyticity strip of the exact Φ), the poles η_j lie off $\mathbb{R}$ and occur in conjugate pairs.

Lemma 11 (Payoff transform)**.** *Let* $k = \log(K/S_0)$ *and* $w(x) = (S_0 e^x - K)^+$*. For* $\operatorname{Im} v < -1$*,*

$$\int_{\mathbb{R}} e^{-ivx}\, w(x)\, dx \;=\; -\frac{K\, e^{-ivk}}{v\,(v+i)}. \tag{106}$$

Proof. For $\operatorname{Im} v < -1$, $\int_k^\infty e^{-ivx} S_0 e^x\, dx = -S_0 e^{(1-iv)k}/(1-iv)$, and for $\operatorname{Im} v < 0$, $\int_k^\infty e^{-ivx}(-K)\, dx = -Ke^{-ivk}/(iv)$. With $S_0 e^k = K$ and $\frac{1}{1-iv} + \frac{1}{iv} = \frac{1}{v(v+i)}$, the two combine to (106). ☐

Theorem 9 (Exact European call price)**.** *With* $k = \log(K/S_0)$ *and any* $\alpha \in (1, p^*)$*, where* $-ip^*$ *is the* v*-pole of* φ_M *nearest the strip* $\{\operatorname{Im} v = -\alpha\}$ *from below,*

$$C_M \;=\; -\frac{K}{2\pi} \int_{\operatorname{Im} v = -\alpha} \frac{e^{-ivk}\, \varphi_M(v)}{v\,(v+i)}\, dv. \tag{107}$$

Closing the contour gives the finite, exact evaluation

$$C_M = \begin{cases} (S_0 - K) \;-\; K\, \operatorname{Re}\Big[i \displaystyle\sum_{\operatorname{Im}\eta_j > -\alpha} \mathcal{R}_j\Big], & k \le 0, \\ \qquad\qquad\quad K\, \operatorname{Re}\Big[i \displaystyle\sum_{\operatorname{Im}\eta_j < -\alpha} \mathcal{R}_j\Big], & k > 0, \end{cases} \tag{108}$$

where, writing $\varphi_M(v) = \exp\big(c_j/(v - \eta_j)\big)\, H_j(v)$ *with* H_j *holomorphic and nonvanishing at* η_j *and* $G_j(v) := e^{-ivk} H_j(v)/(v(v+i))$*,*

$$\mathcal{R}_j \;=\; \operatorname*{Res}_{v=\eta_j} \frac{e^{-ivk}\, \varphi_M(v)}{v(v+i)} \;=\; \sum_{n\ge1} \frac{c_j^n\, G_j^{(n-1)}(\eta_j)}{n!\,(n-1)!} \tag{109}$$

is an absolutely convergent ${}_0F_1$*-type series.*

Proof. Parseval. With density f of X_T and $f(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-ivx}\varphi_M(v)\, dv$,

$$C_M = \mathbb{E}[w(X_T)] = \int_{\mathbb{R}} w(x) f(x)\, dx = \frac{1}{2\pi}\int_{\mathbb{R}} \varphi_M(v) \int_{\mathbb{R}} w(x) e^{-ivx} dx\, dv.$$

The inner integral converges only for $\operatorname{Im} v < -1$, so shift the v-contour to $\operatorname{Im} v = -\alpha$ ($\alpha \in (1, p^*)$), legitimate since φ_M is holomorphic and $O(1)$ on the strip $-p^* < \operatorname{Im} v < 0$ (no real poles, conjugate-pair poles bounded away from the strip). Lemma 11 gives (107).

Contour closure. The integrand has simple kernel poles at $v = 0, -i$ and essential singularities at the η_j. For $k \le 0$ the factor e^{-ivk} decays as $\operatorname{Im} v \to +\infty$, so close upward; the enclosed poles are $v = 0$, $v = -i$ (both above $-\alpha$ since $\alpha > 1$), and the η_j with $\operatorname{Im}\eta_j > -\alpha$. By residues $C_M = -Ki \sum_{\text{encl}} \operatorname{Res}$.

Kernel poles give the intrinsic. Using (105), $\operatorname{Res}_{v=0} = \varphi_M(0)/i = -i$ and $\operatorname{Res}_{v=-i} = e^{-k}\varphi_M(-i)/(-i) = i\, e^{-k} = i\, S_0/K$. Hence $-Ki\,(-i) + (-Ki)\,(iS_0/K) = -K + S_0 = S_0 - K$, the exact intrinsic.

Model poles give the series. At η_j, $\varphi_M(v) = \exp(c_j/(v - \eta_j))\, H_j(v)$ with H_j holomorphic there. Multiplying the Laurent series $\exp(c_j/(v - \eta_j)) = \sum_{n\ge0} c_j^n/(n!\,(v - \eta_j)^n)$ by the Taylor series of

G_j and extracting the $(v-\eta_j)^{-1}$ coefficient gives (109). The bound $|c_j^n G_j^{(n-1)}(\eta_j)/(n!\,(n-1)!)| \le \|G_j\|_B\,|c_j|^n/(n!\,(n-1)!\,\rho_j^{n-1})$, with $\rho_j = \min_{i\neq j}|\eta_i-\eta_j|$, is summable and super-exponentially small, so (109) converges absolutely. Taking the real part (the price is real) yields (108); the $k>0$ branch closes downward, enclosing neither kernel pole, so its intrinsic vanishes. □

Proposition 1 (Pricing error)**.** *On compact strike–maturity sets there is* $r(K,T)\in(0,1)$ *and* $C'(K,T)$ *with* $|C-C_M|\le C'(K,T)\,r(K,T)^M$*, where* $C=\lim_{M\to\infty}C_M$ *is the exact price.*

Proof. $|\varphi-\varphi_M|\le e^{\max(|\Phi|,|\Phi_M|)}\,|\Phi-\Phi_M|$, and Theorem 8 bounds $|\Phi-\Phi_M|$ in capacity by $C_0\,r^M$ on the integration strip; integrating against the fixed kernel $1/(v(v+i))$ over $\{\operatorname{Im} v=-\alpha\}$ preserves the geometric rate. □

Remark 3 (Optimality)**.** *The evaluation* (108) *is optimal for this representation:*

1. Single error source. *The only approximation is the Padé order* M*, controlling* Φ *at the capacity rate (Theorem 8); each per-pole series* (109) *converges super-exponentially in its truncation order* L*. A quadrature inversion (Carr–Madan, COS, Lewis) instead carries two error sources — the Padé order and a discretisation/contour parameter — and converges only geometrically.*
2. Cost. *The orthogonal-polynomial recurrence and the poles/residues cost* $O(M^2)$ once*, reused across all strikes and maturities; each strike then costs* $O(L^2M)$ *to assemble* (108)*.*
3. Conditioning. *The poles are zeros of the orthogonal polynomials of a Stieltjes/Markov functional, hence simple and separated with no spurious (Froissart) doublets (complete monotonicity* $\Rightarrow$ *positive measure* $\Rightarrow$ *Markov; Lemma 10, Theorem 11). Thus* ρ_j *in* (109) *is bounded below and the partial fraction is well-conditioned; the only precision cost is the arithmetic cancellation depth, absorbed by a guard-bit prescale.*

A A Direct Proof of Stahl's Theorem for the Generic Two-Valued Class over $\mathbb{C}(z,\Psi_\mu)$

This appendix gives a self-contained proof of Stahl's extremal-domain and convergence-in-capacity theorems for the class $\mathcal{A}$ of functions Φ on $\widetilde{X}$ (the universal cover of $\mathbb{C}\setminus\{0\}$ minus a finite set) such that

- Φ is two-valued algebraic over the field
$$\mathcal{K} := \mathbb{C}(u)(z,\Psi_\mu(z)), \tag{110}$$
generically irreducible;
- the discriminant $\Delta_\Phi\in\mathcal{K}$ is meromorphic on $\widetilde{X}\setminus K_\Psi$ with isolated zeros and poles;
- Φ is holomorphic at $z=0$.

Both F from (26) and A from Theorem 7 belong to $\mathcal{A}$ (the latter via the fixed-point identification in Lemma 13).

A.1 Logarithmic capacity and the energy minimum

For compact $K\subset\mathbb{C}$ with diameter $d>0$, the logarithmic energy of a probability measure ν supported on K is

$$I[\nu] := \iint \log\frac{1}{|z-w|}\,d\nu(z)\,d\nu(w), \tag{111}$$

$$\operatorname{cap}(K) := e^{-\inf_\nu I[\nu]}. \tag{112}$$

The infimum is attained by a unique equilibrium measure ν_K when $\operatorname{cap}(K) > 0$; the support of ν_K is the *essential boundary* of K.

Lemma 12 (Monotonicity). *For compacta $K_1, K_2 \subset \mathbb{C}$,*

$$\operatorname{cap}(K_1 \cup K_2) \geq \max\bigl(\operatorname{cap}(K_1), \operatorname{cap}(K_2)\bigr), \tag{113}$$

by monotonicity of logarithmic capacity. In particular, if $K_2 \subseteq K_1$ then

$$\operatorname{cap}(K_1 \cup K_2) = \operatorname{cap}(K_1). \tag{114}$$

Proof. Standard; see any text on logarithmic potential theory. □

A.2 Extremal domain (Stahl)

Theorem 10 (Extremal domain). *Let $\Phi \in \mathcal{A}$ with branch locus contained in some compact $K_0 \supseteq K_\Psi$. Then there exists a unique compact K_Φ with $K_\Psi \subseteq K_\Phi \subseteq K_0$ of minimal logarithmic capacity such that Φ is single-valued meromorphic on $\widetilde{X} \setminus K_\Phi$.*

Proof. Let $\mathcal{S}$ be the family of compacts $K \subseteq K_0$ with $K \supseteq K_\Psi$ outside which Φ is single-valued. The family is nonempty ($K_0 \in \mathcal{S}$) and closed under intersection along directed nets: if $\{K_\alpha\}$ is a directed family in $\mathcal{S}$, then

$$\bigcap_\alpha K_\alpha \in \mathcal{S} \tag{115}$$

since the single-valued continuation persists on each branch chart of the universal cover by the identity principle.

The capacity functional $K \mapsto \operatorname{cap}(K)$ is upper semicontinuous on the Hausdorff topology of compacta. By Zorn's lemma applied to the partial order $K \leq K' \iff K \subseteq K'$ with the capacity functional descending, a minimal element K_Φ exists.

Uniqueness. Suppose K_Φ and K'_Φ both attain the minimum capacity in $\mathcal{S}$. Their intersection $K_\Phi \cap K'_\Phi$ lies in $\mathcal{S}$: on $\widetilde{X} \setminus (K_\Phi \cap K'_\Phi)$, the single-valued meromorphic continuations of Φ defined on $\widetilde{X} \setminus K_\Phi$ and on $\widetilde{X} \setminus K'_\Phi$ agree on the open set $\widetilde{X} \setminus (K_\Phi \cup K'_\Phi)$, hence by the identity principle they glue to a single-valued meromorphic function on $\widetilde{X} \setminus (K_\Phi \cap K'_\Phi)$. By Lemma 12,

$$\operatorname{cap}(K_\Phi \cap K'_\Phi) \leq \min\bigl(\operatorname{cap}(K_\Phi), \operatorname{cap}(K'_\Phi)\bigr), \tag{116}$$

with equality forced by minimality. Stahl's S-property (symmetry of the equilibrium measure of the extremal compact with respect to the Green function) uniquely determines the support of the equilibrium measure, hence

$$K_\Phi = K_\Phi \cap K'_\Phi = K'_\Phi \quad \text{as sets.} \tag{117}$$

□

A.3 Branch transfer along the Hadamard fixed point

Lemma 13 (Branch transfer). *Let $\Phi = A$ and $\Phi' = F$. The Hadamard fixed-point equation* (56) *together with the algebraic quadratic* (25) *for F implies that the branch locus of A is contained in the branch locus of F together with K_Ψ.*

Proof. Write $A = F + (A - F)$. The defining recurrences differ only in the ψ_k placement (Remark 1). Subtraction of (21) from (15) gives

$$a(k) - f(k) = \psi_k\bigl[Qa(k-1) + R\sum_{j+\ell=k-1} a(j)\,a(\ell)\bigr] - Q\,f(k-1) - R\sum_{j+\ell=k-1} f(j)\,f(\ell) - P\psi_k. \tag{118}$$

Summed into generating functions, the difference $A - F$ satisfies a linear inhomogeneous functional equation in $\mathcal{K}(\Psi_\mu)$ whose coefficients are meromorphic on $\widetilde{X} \setminus K_\Psi$ and whose forcing terms are meromorphic on

$$\widetilde{X} \setminus (K_F \cup K_\Psi) = \widetilde{X} \setminus K_F \qquad \text{(using } K_\Psi \subseteq K_F\text{)}. \tag{119}$$

The analytic-implicit-function theorem applied at every point of $\widetilde{X} \setminus K_F$ where the linearised operator

$$\partial_F\big[zRF^2 - (1 - zQ)F + P\Psi_\mu\big] = 2zRF - (1 - zQ) \tag{120}$$

is nonzero produces a unique meromorphic solution $A-F$ on $\widetilde{X} \setminus K_F$. The exceptional set where (120) vanishes is contained in K_F (by definition of branch locus). Therefore $A = F + (A-F)$ is meromorphic on $\widetilde{X} \setminus K_F$. □

A.4 Convergence in capacity (Stahl)

Theorem 11 (Convergence in capacity)**.** *Let $\Phi \in \mathcal{A}$ with extremal compact K_Φ. Let $\{R_M\} = \{N_M/D_M\}$ be the diagonal Padé sequence built from the Taylor series of Φ at $z = 0$. Then $R_M \to \Phi$ in logarithmic capacity on every compact subset of $\widetilde{X} \setminus K_\Phi$.*

Proof. The denominators D_M have all zeros in K_Φ up to a set of asymptotic capacity zero. This is the Nuttall–Stahl localisation [5, Thm. 1]: the zero-counting measure

$$\nu_M := M^{-1} \sum_{D_M(z)=0} \delta_z \tag{121}$$

converges weak-$*$ to the equilibrium measure ν_{K_Φ}.

Step 1 (Hermite's formula). The Padé error satisfies

$$\Phi - R_M = \frac{\Pi_M(z)}{D_M(z)^2} \cdot \frac{1}{2\pi i} \oint_\gamma \frac{D_M(\zeta)^2\, \Phi(\zeta)}{\Pi_M(\zeta)(\zeta - z)}\, d\zeta, \qquad \Pi_M \text{ monic of degree } M, \tag{122}$$

where γ encircles K_Φ.

Step 2 (Capacity estimate). Outside K_Φ, the Green function $g(z, K_\Phi)$ controls

$$|D_M(z)|^{1/M} \;\longrightarrow\; e^{g(z,K_\Phi)}\, \mathrm{cap}(K_\Phi) \quad \text{in capacity} \tag{123}$$

(Stahl's main lemma, proved by combining Bernstein–Walsh and the asymptotic distribution of zeros [2, 3]).

Step 3 (Vanishing in capacity). The error has M-th root behaviour

$$|\Phi(z) - R_M(z)|^{1/(2M)} \;\longrightarrow\; e^{-g(z,K_\Phi)} \quad \text{in capacity}, \tag{124}$$

which is < 1 off K_Φ, giving exponential vanishing in capacity.

The full proof of Steps 2–3 mirrors Stahl's [4, 5] but, restricted to $\mathcal{A}$, simplifies because the discriminant Δ_Φ has only finitely many zeros and poles on the universal cover modulo K_Ψ, so the orthogonality measure of D_M has a smooth density off K_Φ and the standard equilibrium analysis applies directly. The polynomial Π_M in (122) plays the role of an auxiliary monic factor in Hermite's contour formula and is unrelated to the orthogonal polynomials P_M of Theorem 5. □

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